

Simpler Presentations for Many Fragments of Quantum Circuits

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Abstract

Equational reasoning is central to quantum circuit optimisation and verification: one replaces subcircuits by provably equivalent ones using a fixed set of rewrite rules viewed as equations. We study such reasoning through finite equational theories, presenting restricted quantum gate fragments as symmetric monoidal categories (PROPs), where wire permutations are treated as structural and separated cleanly from fragment-specific gate axioms. For six widely used near-Clifford fragments: qubit Clifford, real Clifford, Clifford+ T (up to two qubits), Clifford+ CS (up to three qubits), CNOT-dihedral, we transfer the completeness results of prior work into our PROP framework. Beyond completeness, we address minimality (axiom independence). Using uniform separating interpretations into simple semantic targets, we prove minimality for several fragments (including all arities for qubit Clifford, real Clifford, and CNOT-dihedral), and bounded minimality for the remaining cases. Overall, our presentations significantly reduce rule counts compared to prior work and provide a reusable categorical framework for constructing complete and often minimal rewrite systems for quantum circuit fragments.

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1 Introduction

Quantum circuit optimisation and verification rely on *equational reasoning*: replacing subcircuits by provably equivalent ones. We study such reasoning via finite equational theories (rewrite rules viewed as equations) for restricted gate sets. A rule set is *complete* if it derives every equality between circuits denoting the same unitary [5, 4].

Categorically, circuits form morphisms in a symmetric monoidal category, and a finite equational theory presents the corresponding symmetric monoidal subcategory as a PROP by generators and relations [6, 10]. Besides completeness, we are interested in *minimality* (axiom independence): no rule should be derivable from the others. Minimal axiom sets are conceptually informative and support leaner automated rewriting and proof search.

Context. Complete finite presentations are known for several near-Clifford fragments: Selinger’s stabilizer presentation for n -qubit Clifford operators [11], extensions such as Clifford+ T and Clifford+ CS [2, 3], qutrit Clifford [7], and the CNOT-dihedral fragment [1]. However, these presentations are not minimal, and even identifying irredundant axioms can be subtle. For universal circuit models, finite complete axiomatisations are known to require rules of unbounded arity [4], motivating a systematic study of fragments where finite and sometimes minimal presentations are achievable.



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Contributions. We develop a uniform PROP framework covering six fragments:

qubit Clifford, real Clifford, Clifford+ T , Clifford+ CS , CNOT-dihedral, qutrit Clifford.

For each fragment we provide a finite equational presentation and prove completeness for strict unitary equality. We do this by:

- a conservative scalar-refinement step when needed;
- a uniform passage from PRO presentations to PROPs; and
- explicit encoding/decoding PROP morphisms plus a generic transfer theorem (Theorem 21).

We also prove axiom independence whenever currently achievable, including: qubit Clifford and real-qubit Clifford (all arities), CNOT-dihedral (all arities), Clifford+ T on one qubit, Clifford+ CS on two qubits, and qutrit Clifford on up to two qutrits (Theorem 35). Independence is shown uniformly via separating functors into simple semantic targets (counting monoids, projective semantics, determinant-phase invariants).

Table 1 summarises the resulting axiom counts against previous presentations.

Fragment	Previous Theory	Our Theory	Figure
Qubit Clifford on n qubits	15 rules [11]	8 rules (minimal for n qubits)	1
Real Clifford on n qubits	16 rules [9]	10 rules (minimal for n qubits)	2
Qutrit Clifford on n qutrits	18 rules [7]	10 rules (minimal for 2 qutrits, conjectured for n)	3
Clifford+ T on 2 qubits	18 rules [2]	11 rules (minimal for 1 qubit, simplified for 2)	4
Clifford+ CS on 3 qubits	17 rules [3]	14 rules (minimal for 2 qubits, simplified for 3)	5
CNOT-dihedral on n qubits	13 rules [1]	11 rules (minimal for n qubits)	6

■ **Table 1** Comparison of rule counts: previous work vs. our presentations¹

2 Graphical languages for quantum circuits

This section fixes the categorical syntax and proof notion used throughout the paper. We model circuits as morphisms in strict symmetric monoidal categories with objects the wire counts, and we present circuit fragments by *generators and relations*. The resulting notion of completeness is *faithfulness* of an interpretation functor; we will use this setup verbatim in Sections 3 and 4.

2.1 Graphical languages

Our basic ambient setting is strict (symmetric) monoidal. The object n represents n parallel wires, tensor corresponds to placing circuits side by side, and sequential composition corresponds to plugging outputs into inputs. Strictness is a bookkeeping choice: associativity and unit constraints are treated as literal equalities, so we can draw string diagrams without parentheses.

We use the standard one-object-per-wire formalisms: PROs and PROPs in the sense of Joyal–Street [6, 10]. Operationally, the distinction is that a *PROP* has wire permutations as

¹ Several previous presentations work in a PRO setting where wire permutations are implicit. In our PROP setting these symmetries are structural and not counted as axioms, so raw rule counts are not strictly comparable.

built-in (structural) morphisms, whereas a *PRO* does not. Since our circuit fragments are permutation-invariant, working in a *PROP* matches the intended graphical intuition and cleanly separates *structural* equalities (coherence and swaps) from *gate* equalities (axioms).

► **Definition 1.** A *PRO* is a strict monoidal category $\mathbf{P}$ whose objects are the natural numbers $n \in \mathbb{N}$ and whose tensor on objects is addition $n \otimes m := n + m$ with unit 0. We write $\mathbf{P}(n, m)$ for morphisms $n \rightarrow m$, with sequential composition $\circ$ and parallel composition $\otimes$, subject to the usual strict monoidal axioms (??????????).

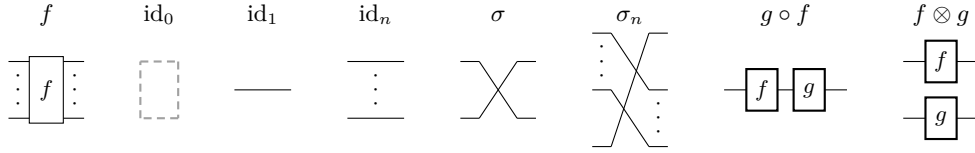
Intuitively, a *PRO* is a typed circuit calculus with n input wires and m output wires, where $\circ$ is plugging and $\otimes$ is juxtaposition, but where wire permutations are not available as structural morphisms.

We write $f : n \rightarrow m$ when $f \in \mathbf{P}(n, m)$; n and m are the numbers of input and output wires of f . In particular $\text{id}_n = \text{id}_1^{\otimes n}$ for all $n \in \mathbb{N}$.

► **Definition 2.** A *PROP* is a *PRO* $\mathbf{P}$ equipped with a symmetric monoidal structure, i.e. symmetry isomorphisms $\sigma_{n,m} : n + m \rightarrow m + n$ ($n, m \in \mathbb{N}$) generated by the basic swap $\sigma := \sigma_{1,1}$ and satisfying the axioms (??)-(??) in ???. We write $\sigma_n := \sigma_{n,1} : n + 1 \rightarrow 1 + n$ for swapping one wire with a block of n wires.

Thus in a *PROP*, swaps and permutations are part of the ambient structure (and hence will not be counted as fragment-specific axioms later on).

PROPs admit a convenient string-diagram representation: a morphism $f : n \rightarrow m$ is drawn as a box with n incoming wires and m outgoing wires. Sequential composition is drawn by stacking boxes, tensor by placing them side by side, and $\sigma_{n,m}$ by crossing a block of n wires with a block of m wires. With these conventions, the coherence axioms are exactly the diagrammatic moves that make deformation sound.



The coherence axioms for *PROs* and *PROPs*, used throughout the paper, are summarised in ??. From now on, equalities obtained solely by these coherence moves (including permutations) are treated as *structural* and used silently in diagrammatic arguments.

2.2 Free *PROPs* and presentations

This paper treats a circuit fragment as: *pick a gate signature, then impose equations*. Categorically, this is a quotient of a free *PROP*. The point of working at this level of generality is that it makes the boundary between structural equalities and fragment-specific equalities completely explicit; we will rely on this separation later, notably when comparing axiom counts across presentations.

Let Σ be a set of generating morphisms equipped with arity functions $\text{dom}, \text{cod} : \Sigma \rightarrow \mathbb{N}$. We write $g : \text{dom}(g) \rightarrow \text{cod}(g)$ for $g \in \Sigma$.

► **Definition 3.** Let Σ be a set of generators equipped with arities $\text{dom}, \text{cod} : \Sigma \rightarrow \mathbb{N}$. The free *PROP* on Σ , written $\mathbf{P}_\Sigma$, is the *PROP* whose morphisms are string diagrams generated from Σ together with the *PROP* structural morphisms (identities and symmetries), modulo *PROP* coherence.

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► **Definition 4.** Given a set $\mathcal{R}$ of well-typed equations between morphisms of $\mathbf{P}_\Sigma$, the presented PROP $\mathbf{P}_\Sigma/\mathcal{R}$ is the quotient of $\mathbf{P}_\Sigma$ by the congruence generated by $\mathcal{R}$.

This quotient is the formal object that we treat as our *graphical language*: diagrams are equal exactly when their equality is derivable from the coherence axioms together with the chosen relations.

► **Remark 5.** We distinguish:

- *structural equality* in $\mathbf{P}_\Sigma$, written $C_1 \equiv_\Sigma C_2$, meaning equality using only PROP coherence (associativity/unit laws and symmetries); and
- *derivability* in a presented PROP $\mathbf{P}_\Sigma/\mathcal{R}$, written $\mathbf{P}_\Sigma/\mathcal{R} \vdash C_1 = C_2$, meaning equality derivable from $\mathcal{R}$ using equational reasoning (plus structural moves).

2.3 Monoidal functors and completeness

To connect a presented syntax $\mathbf{P}_\Sigma/\mathcal{R}$ to its intended meaning, we interpret generators as concrete operations and extend compositionally. Completeness will then assert that the equations generated by $\mathcal{R}$ capture *all* semantic equalities, with no extra identities hiding in the semantics.

► **Definition 6.** A PROP morphism is a strict symmetric monoidal functor $F : \mathbf{P}_1 \rightarrow \mathbf{P}_2$ between PROPs that is the identity on objects. Equivalently, F preserves identities, symmetries, sequential composition and tensor.

► **Definition 7.** Let $\mathbf{P}_\Sigma/\mathcal{R}$ be a presented PROP and let $\mathcal{C}$ be a symmetric monoidal category. An interpretation functor is a strict symmetric monoidal functor $\llbracket \cdot \rrbracket : \mathbf{P}_\Sigma \rightarrow \mathcal{C}$ specified on generators $g \in \Sigma$ and extended monoidally to all diagrams.

If every equation in $\mathcal{R}$ holds in $\mathcal{C}$, then $\llbracket \cdot \rrbracket$ factors uniquely through the quotient, yielding a functor $\llbracket \cdot \rrbracket : \mathbf{P}_\Sigma/\mathcal{R} \rightarrow \mathcal{C}$.

► **Definition 8.** Let $\llbracket \cdot \rrbracket : \mathbf{P}_\Sigma/\mathcal{R} \rightarrow \mathcal{C}$ be such an interpretation. We say that the graphical language $\mathbf{P}_\Sigma/\mathcal{R}$ is complete for this semantics if, for all circuits C_1, C_2 of the same type, $\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket$ implies $\mathbf{P}_\Sigma/\mathcal{R} \vdash C_1 = C_2$. Equivalently, the interpretation functor is faithful.

We will also use a bounded variant. Given $k \in \mathbb{N}$, we say that $\mathbf{P}_\Sigma/\mathcal{R}$ is complete up to k wires if the above implication holds whenever $C_1, C_2 : n \rightarrow m$ with $n, m \leq k$. Equivalently, the interpretation functor is faithful on each hom-set $\mathbf{P}_\Sigma/\mathcal{R}(n, m)$ for $n, m \leq k$.

Operationally, completeness says that semantic equality can always be witnessed by an explicit diagrammatic derivation from the chosen relations; the bounded form restricts this requirement to circuits of limited arity (as will occur for some near-Clifford fragments later).

When $\mathcal{C}$ is (by definition) the symmetric monoidal subcategory of some ambient category generated by the images of the generators, the interpretation functor is also full; in that case $\mathbf{P}_\Sigma/\mathcal{R}$ is actually isomorphic to the semantic subPROP $\mathcal{C}$.

► **Remark 9.** Several completeness results in the literature are stated for PROs (no built-in symmetry). Since we work in PROPs, we record a uniform lifting principle.

Let $\mathbf{P}/\mathcal{R}$ be a complete PRO presentation with a strict monoidal interpretation into a symmetric monoidal category $\mathcal{C}$ (viewed as monoidal by forgetting symmetry). Assume there exists a PRO circuit $\tau : 2 \rightarrow 2$ such that $\llbracket \tau \rrbracket$ is the symmetry $\sigma_{1,1}$ of $\mathcal{C}$.

Form the free PROP on the same generators, and quotient by $\mathcal{R}$ together with the single equation $\sigma_{1,1} = \tau$ (identifying the *structural* swap with the *definable* swap τ). Define a translation T from PROP diagrams to PRO diagrams by recursively replacing each structural

swap $\sigma_{1,1}$ by τ (and leaving generators untouched). Then $\llbracket D \rrbracket = \llbracket T(D) \rrbracket$ for all PROP diagrams D and any semantic equality between PROP diagrams reduces to a semantic equality between their PRO translations. By PRO completeness, $T(D_1) = T(D_2)$ is derivable from $\mathcal{R}$, and the added equation $\sigma_{1,1} = \tau$ lifts this derivation back to the PROP setting. Hence completeness transfers.

3 Quantum circuits and their relations

This section fixes the concrete syntax and semantics used throughout the paper. We define a uniform “master” setting for qubit circuits and a direct setting for qutrit Clifford circuits, and we package each of our six fragments as a presented subPROP together with its intended unitary interpretation. These definitions are the input to the completeness transfer of Section 4 and to the independence arguments of Section 5.

For qubits we use a single master signature containing all gates needed across the qubit fragments; each fragment is obtained by restriction to a sub-signature and by imposing a finite rule set. For qutrits we only need the Clifford fragment, so we present it directly.

► **Definition 10.** *The qubit master signature $\Sigma^{(2)}$ consists of the following non-structural generators:*

$$\omega, \ominus : 0 \rightarrow 0, \quad \boxed{\text{H}}, \boxed{\text{Z}}, \boxed{\text{S}}, \boxed{\text{T}}, \boxed{\text{X}} : 1 \rightarrow 1, \quad \text{CNOT}, \text{SWAP} : 2 \rightarrow 2.$$

We write $\mathbf{P}^{(2)} := \mathbf{P}_{\Sigma^{(2)}}$ for the free PROP on $\Sigma^{(2)}$. Structural morphisms id_n and $\sigma_{n,m}$ come from the ambient PROP structure and are not counted as generators.

Intuitively, $\Sigma^{(2)}$ is a “common super-signature”: each qubit fragment later selects a sub-signature $\Sigma_\bullet \subseteq \Sigma^{(2)}$ and adds fragment-specific equations.

Note that every generator $g \in \Sigma^{(2)}$ satisfies $\text{dom}(g) = \text{cod}(g)$. Hence every well-typed diagram in $\mathbf{P}^{(2)}$ has type $n \rightarrow n$ for some n , and equational derivations never change the wire count. We exploit this arity preservation in Section 5.

Let $\mathbf{FdHilb}$ be the symmetric monoidal category of finite-dimensional complex Hilbert spaces and linear maps, with tensor product $\otimes$, tensor unit $\mathbb{C}$, and symmetry given by swapping tensor factors. By Mac Lane’s coherence theorem [8] we may regard $\mathbf{FdHilb}$ as strict monoidal.

► **Definition 11.** *Define **Qubit** to be the PROP with objects $\mathbb{N}$ and hom-sets*

$$\mathbf{Qubit}(n, m) := \begin{cases} \{U : (\mathbb{C}^2)^{\otimes n} \rightarrow (\mathbb{C}^2)^{\otimes n} \mid U \text{ unitary}\} & n = m, \\ \emptyset & n \neq m, \end{cases}$$

*with composition, identities, tensor, and symmetry inherited from $\mathbf{FdHilb}$ (viewed as strict monoidal). Define **Qutrit** similarly, replacing $\mathbb{C}^2$ by $\mathbb{C}^3$.*

► **Remark 12.** Both **Qubit** and **Qutrit** are endomorphism-only: there are no unitaries $n \rightarrow m$ for $n \neq m$.

► **Definition 13.** *For any PROP $\mathbf{P}$ and any set G of morphisms, we write $\langle G \rangle$ for the subPROP generated by G (with the same objects as $\mathbf{P}$).*

Operationally, $\langle G \rangle$ is the smallest PROP containing all structural morphisms and the chosen generators G , and closed under $\circ$ and $\otimes$.

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By the universal property of the free PROP, there is a unique strict symmetric monoidal functor $\llbracket \cdot \rrbracket^{(2)} : \mathbf{P}^{(2)} \longrightarrow \mathbf{Qubit}$ sending:

$$\begin{array}{ll} \llbracket \omega \rrbracket^{(2)} : 1 \mapsto e^{i\pi/4} & \llbracket \text{H} \rrbracket^{(2)} : |x\rangle \mapsto \frac{1}{\sqrt{2}} \sum_{k=0}^1 (-1)^{xk} |k\rangle \\ \llbracket \ominus \rrbracket^{(2)} : 1 \mapsto -1 & \llbracket \text{S} \rrbracket^{(2)} : |x\rangle \mapsto i^x |x\rangle \\ \llbracket \text{X} \rrbracket^{(2)} : |x\rangle \mapsto |1-x\rangle & \llbracket \text{Z} \rrbracket^{(2)} : |x\rangle \mapsto (-1)^x |x\rangle \\ \llbracket \text{CNOT} \rrbracket^{(2)} : |x, y\rangle \mapsto |x, x \oplus y\rangle & \llbracket \text{T} \rrbracket^{(2)} : |x\rangle \mapsto e^{i\pi x/4} |x\rangle \\ \llbracket \text{CNOT} \rrbracket^{(2)} : |x, y\rangle \mapsto i^{xy} |x, y\rangle & \end{array}$$

for $x, y \in \{0, 1\}$ and addition $\oplus$ modulo 2.

3.1 Qubit circuit fragments as presented subPROPs

We now package each qubit fragment uniformly as a quotient of a free PROP on a sub-signature of $\Sigma^{(2)}$, together with its induced unitary semantics. This gives a common interface for later arguments: completeness will be stated as faithfulness of the resulting interpretation functors, and minimality will be phrased as independence inside these presented PROPs.

All qubit fragments studied in this paper are interpreted as subPROPs of **Qubit** and are presented as quotients of subPROPs of $\mathbf{P}^{(2)}$. As explained in Section 2.2, each fragment is determined by a sub-signature $\Sigma_\bullet \subseteq \Sigma^{(2)}$ and a finite set of relations $C_\bullet$ on the free PROP $\mathbf{P}_{\Sigma_\bullet}$. The resulting quotient PROP is our formal circuit language for the fragment.

Throughout the rest of this subsection, the placeholder $\bullet$ ranges over the *five qubit fragments* considered here: $\bullet \in \{\text{Cliff}, \text{RCliff}, \text{CliffT}, \text{CliffCS}, \text{CNOTdihe}\}$.

► **Definition 14.** Fix a sub-signature $\Sigma_\bullet \subseteq \Sigma^{(2)}$. Let $\mathbf{P}_\bullet := \mathbf{P}_{\Sigma_\bullet}$ be the free PROP on $\Sigma_\bullet$, and (by the universal property of the free PROP) identify it with its canonical image in $\mathbf{P}^{(2)}$.

A finite set $C_\bullet$ of well-typed equations between morphisms of $\mathbf{P}_\bullet$ will be called a rule set (or axiom set) for the fragment $\bullet$. We write $\mathbf{QCirc}_\bullet := \mathbf{P}_\bullet / C_\bullet$ for the corresponding presented PROP (circuits modulo the axioms).

Thus $\mathbf{QCirc}_\bullet$ is the syntax of the fragment: morphisms are circuits built from $\Sigma_\bullet$, quotiented by the chosen equations $C_\bullet$ (with PROP structure and symmetries treated as structural).

For each fragment $\bullet$ we first fix its intended unitary semantics as the subPROP $\mathbf{Qubit}_\bullet := \langle \llbracket g \rrbracket^{(2)} \mid g \in \Sigma_\bullet \rangle \subseteq \mathbf{Qubit}$, generated by the interpretations of the gates in $\Sigma_\bullet$; equivalently, this is the intended semantic category of the fragment. Since the master interpretation $\llbracket \cdot \rrbracket^{(2)}$ restricts to $\mathbf{P}_\bullet$ and lands in $\mathbf{Qubit}_\bullet$, and since every equation of $C_\bullet$ holds semantically, it factors uniquely through the quotient to a well-defined strict symmetric monoidal functor $\llbracket \cdot \rrbracket_\bullet : \mathbf{QCirc}_\bullet \rightarrow \mathbf{Qubit}_\bullet$.

When $\llbracket \cdot \rrbracket_\bullet$ is faithful we say that the presentation $\mathbf{QCirc}_\bullet$ is *complete* for the corresponding fragment (in the sense of Definition 8).

We now instantiate this scheme for each qubit fragment. For compactness, we collect the defining data (generators and rule sets) in a single definition.

► **Definition 15.** The five qubit fragments treated in this paper are specified by the following sub-signatures and rule sets:

Fragment	$\Sigma_\bullet \subseteq \Sigma^{(2)}$	Rule set $C_\bullet$
Clifford	$\Sigma_{Cliff} := \{ \omega, \text{H}, \text{S}, \text{CNOT} \}$	$C_{Cliff} := \text{Cliff}_2$ (Figure 1)
Real Clifford	$\Sigma_{RCliff} := \{ \omega, \text{H}, \text{Z}, \text{CNOT} \}$	$C_{RCliff} := \text{RCliff}_2$ (Figure 2)
Clifford+T	$\Sigma_{CliffT} := \{ \omega, \text{H}, \text{T}, \text{CNOT} \}$	$C_{CliffT} := \text{CliffT}_2$ (Figure 4)
Clifford+CS	$\Sigma_{CliffCS} := \{ \omega, \text{H}, \text{S}, \text{CNOT} \}$	$C_{CliffCS} := \text{CliffCS}_2$ (Figure 5)
CNOT-dihedral	$\Sigma_{CNOTdihe} := \{ \omega, \text{X}, \text{T}, \text{CNOT} \}$	$C_{CNOTdihe} := \text{CNOTdihe}_2$ (Figure 6)

These data determine the five presented PROPs $\mathbf{QCirc}_\bullet$ (cf. Definition 14). For each fragment $\bullet$, restricting the fixed interpretation $\llbracket \cdot \rrbracket^{(2)}$ yields the PROP morphism $\llbracket \cdot \rrbracket_\bullet : \mathbf{QCirc}_\bullet \rightarrow \mathbf{Qubit}_\bullet$ defined above.

3.2 The qutrit Clifford fragment

On the qutrit side we only need the Clifford fragment. Since no other qutrit gate families are considered in this work, we present qutrit Clifford circuits directly rather than introducing a separate qutrit master signature. We will use this presentation exactly as stated in the completeness transfer of Section 4.

For qutrits we use the computational basis $\{|0\rangle, |1\rangle, |2\rangle\}$ of $\mathbb{C}^3$, and the induced basis of $(\mathbb{C}^3)^{\otimes n}$.

► **Definition 16.** The qutrit Clifford fragment is generated by the signature

$$\Sigma_{Cliff3} := \{ \omega : 0 \rightarrow 0, \text{H} : 1 \rightarrow 1, \text{S} : 1 \rightarrow 1, \text{CNOT} : 2 \rightarrow 2 \}.$$

Let $\mathbf{P}_{Cliff3} := \mathbf{P}_{\Sigma_{Cliff3}}$ be the free PROP on Σ_{Cliff3} . Let $C_{Cliff3} := \text{Cliff}_3$ be the rule set given in Figure 3. We then define $\mathbf{QCirc}_{Cliff3} := \mathbf{P}_{Cliff3} / C_{Cliff3}$.

Thus $\mathbf{QCirc}_{Cliff3}$ is the PROP of qutrit Clifford circuits modulo the axioms C_{Cliff3} (with swaps treated as structural, as usual in a PROP).

By the universal property of the free PROP, there is a unique strict symmetric monoidal functor $\llbracket \cdot \rrbracket_{Cliff3} : \mathbf{QCirc}_{Cliff3} \rightarrow \mathbf{Qutrit}_{Cliff3}$ determined by

$$\begin{aligned} \llbracket \omega \rrbracket_{Cliff3} : 1 &\mapsto e^{i\pi/6} & \llbracket \text{H} \rrbracket_{Cliff3} : |x\rangle &\mapsto \frac{1}{\sqrt{3}} \sum_{k=0}^2 e^{2\pi i x k/3} |k\rangle \\ \llbracket \text{CNOT} \rrbracket_{Cliff3} : |x, y\rangle &\mapsto |x, (x+y) \bmod 3\rangle & \llbracket \text{S} \rrbracket_{Cliff3} : |x\rangle &\mapsto e^{i\pi x(x-1)/3} |x\rangle \end{aligned}$$

for $x, y \in \{0, 1, 2\}$. Here $\mathbf{Qutrit}_{Cliff3} := \langle \llbracket g \rrbracket_{Cliff3} \mid g \in \Sigma_{Cliff3} \rangle$ is the semantic subPROP generated by the qutrit Clifford gates.

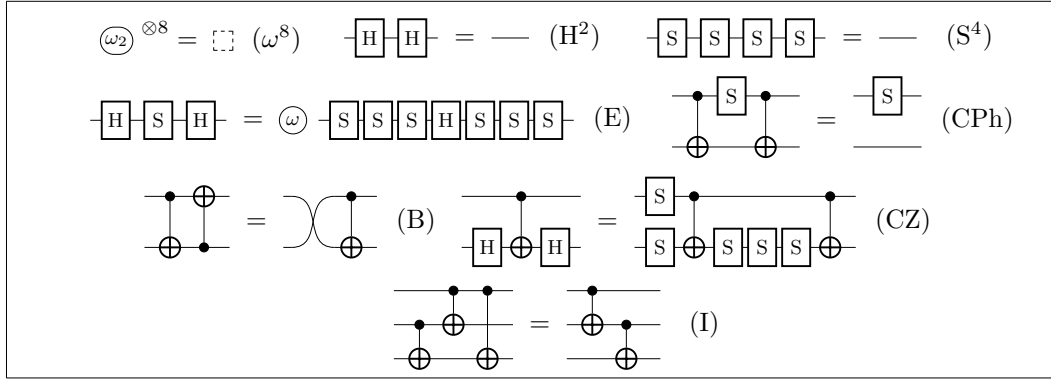
4 Completeness: from PROs to PROPs

This section explains how we obtain completeness of the presentations from Section 3 without re-proving fragment-specific normal-form theorems. For each fragment, we transfer completeness from an existing sound and complete presentation in the literature [11, 9, 7, 2, 3, 1]. The transfer is packaged as a uniform pipeline that we will reuse across all six fragments; all fragment-dependent verifications are deferred to the cited appendices.

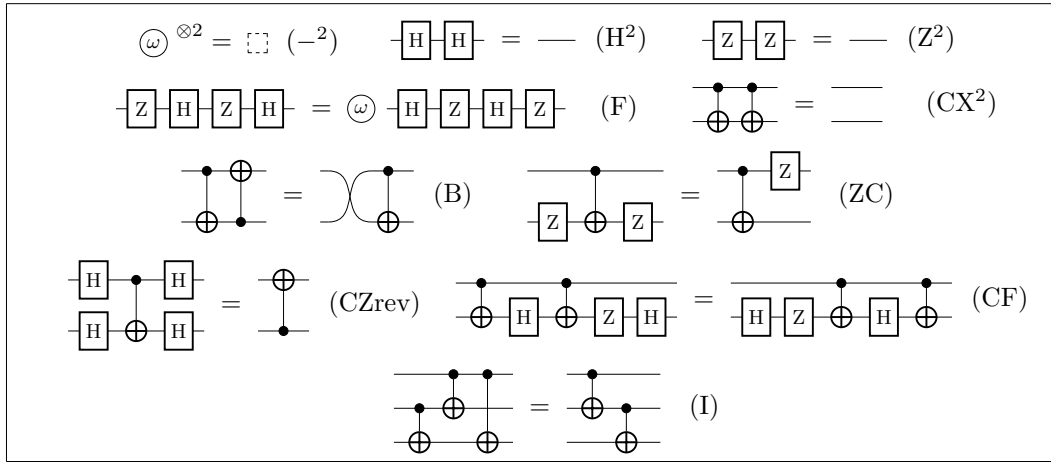
The pipeline has three conceptual steps:

1. **PRO** $\rightarrow$ **PROP**. If the source theory is stated for a PRO, we first pass to a PROP presentation using Remark 9.

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■ **Figure 1** Set of relations Cliff_2 .



■ **Figure 2** Set of relations RCliff_2 .

2. **Scalar alignment.** When source and target use different scalar conventions, we perform a *scalar refinement* so that both syntaxes interpret into the same semantic subPROP (Section 4.1).
3. **Encoding/decoding.** We relate the two syntaxes by explicit encoding/decoding PROP morphisms and apply a generic transfer theorem (Theorem 21).

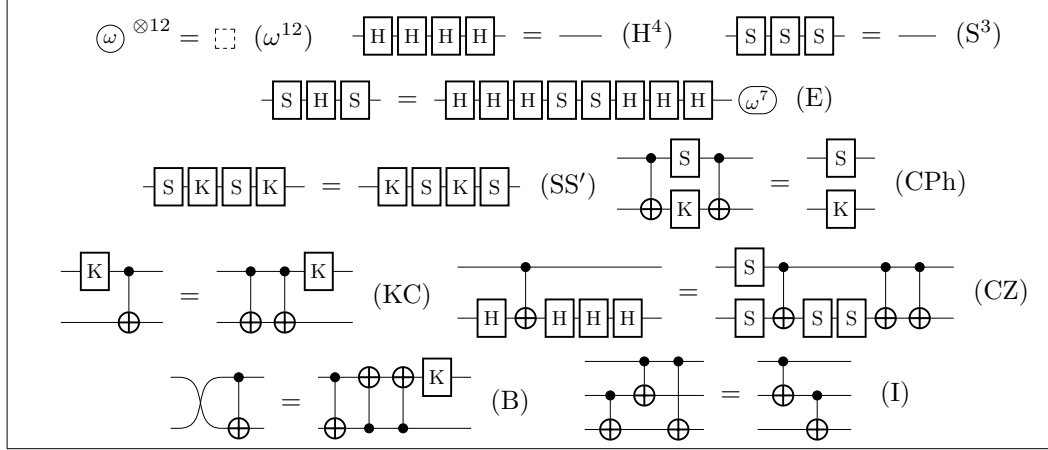
Throughout, “complete” means *faithful* in the sense of Def. 8, for the interpretation functors fixed in Section 3.

From now on, whenever a source completeness theorem is stated for a PRO, we tacitly pass to the corresponding PROP using Remark 9. Thus all translation functors considered below are strict symmetric monoidal functors and are the identity on objects.

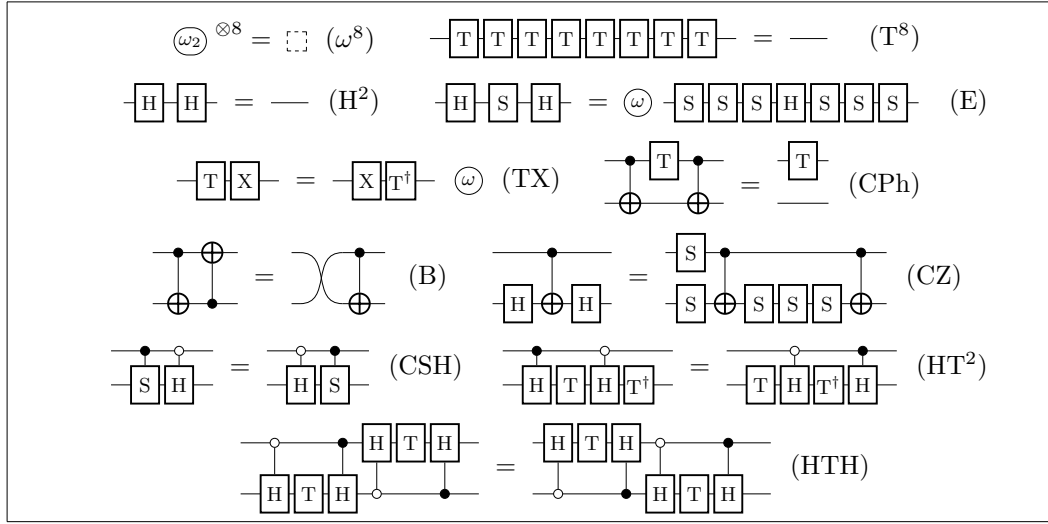
4.1 Scalar refinement

► **Definition 17.** For any PROP $\mathcal{C}$, its scalars are the endomorphisms of the tensor unit, i.e. the hom-set $S(\mathcal{C}) := \mathcal{C}(0, 0)$.

Before comparing presentations by encoding/decoding, source and target must interpret into the *same* semantic subPROP. In our setting we work with *strict* unitary equality, so



■ **Figure 3** Set of relations Cliff_3 , where the non-generator gates are defined in ??.



■ **Figure 4** Set of relations CliffT_2 , where the non-generator gates are defined in ??.

global phases are not quotiented away: each fragment comes with a distinguished finite scalar subgroup $S(\mathcal{C})$, and matching scalar conventions is part of matching the semantics.

Two of the imported completeness results are stated with a *smaller* scalar subgroup than the one fixed in Section 3. Concretely:

- the qutrit Clifford presentation of [7] uses an order-6 scalar group, whereas our qutrit semantics uses an order-12 scalar group; and
- the Clifford+ CS presentation of [3] uses an order-4 scalar group, whereas our qubit setting uses an order-8 scalar group.

We therefore apply the scalar-refinement construction of Appendix 7 to *conservatively* enlarge the scalar subgroup of these source presentations, so that they can be regarded as complete presentations for the same gate fragment in the same strict semantics as our target PROPs. By Lemma 41, this refinement preserves faithfulness: it introduces no new equalities between non-scalar circuits, and only adds the missing global phases.

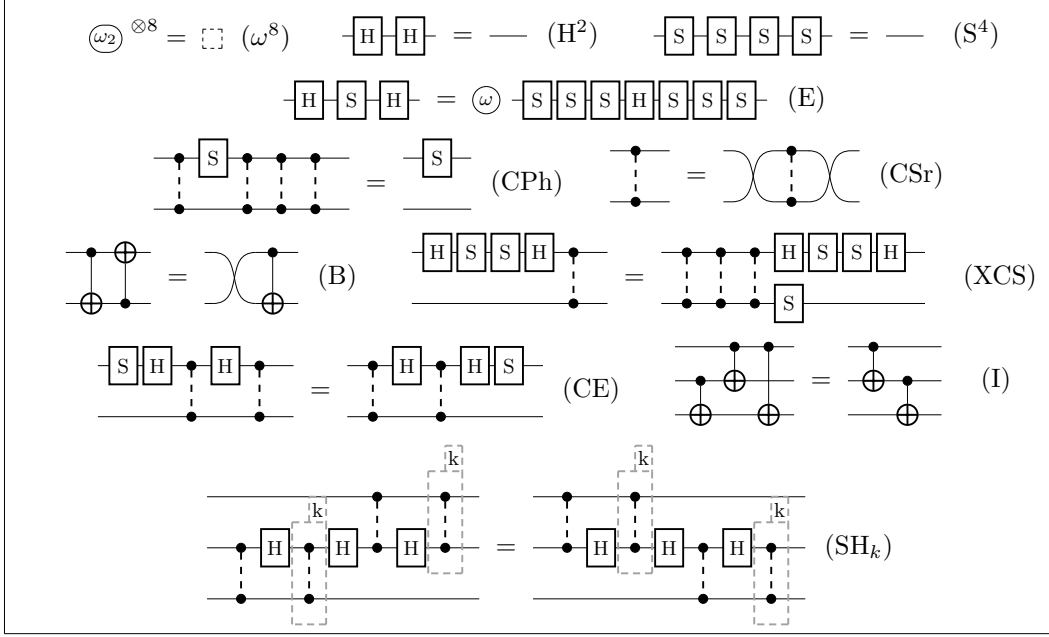


Figure 5 Set of relations CliffCS_2 , where the non-generator gates are defined in ???. In Equation (SH_k) , the dashed box is a *repetition box* indexed by $k \in \{0, 1, 2, 3\}$. Concretely, when $k = 0$ the subcircuit inside the box is omitted; when $k = 1$ it appears once; when $k = 2$ it appears twice (sequentially); and when $k = 3$ it appears three times. Hence Equation (SH_k) actually unfolds into four separate axioms, one for each value of k .

4.2 Encoding, decoding, and transfer of completeness

We now state the abstract transfer argument used repeatedly in the rest of the paper. The setup is: a *source* presentation is already known complete, and we have a *target* presentation for the same semantics. We prove completeness of the target presentation by exhibiting translations in both directions that preserve semantics and are inverse up to derivability.

Let $\mathbf{P}^{\text{src}}/\mathcal{R}^{\text{src}}$ be a complete presentation with interpretation into $\mathcal{C}^{\text{src}} \subseteq \mathbf{FdHilb}$, and let $\mathbf{P}/\mathcal{R}$ be a target presentation interpreted by $\llbracket \cdot \rrbracket : \mathbf{P}/\mathcal{R} \rightarrow \mathcal{C} \subseteq \mathbf{FdHilb}$.

Definition 18. A pair of strict symmetric monoidal functors $E : \mathbf{P} \rightarrow \mathbf{P}^{\text{src}}$, $D : \mathbf{P}^{\text{src}} \rightarrow \mathbf{P}$ is an encoding/decoding pair if, for all circuits $C \in \mathbf{P}$ and $C' \in \mathbf{P}^{\text{src}}$,

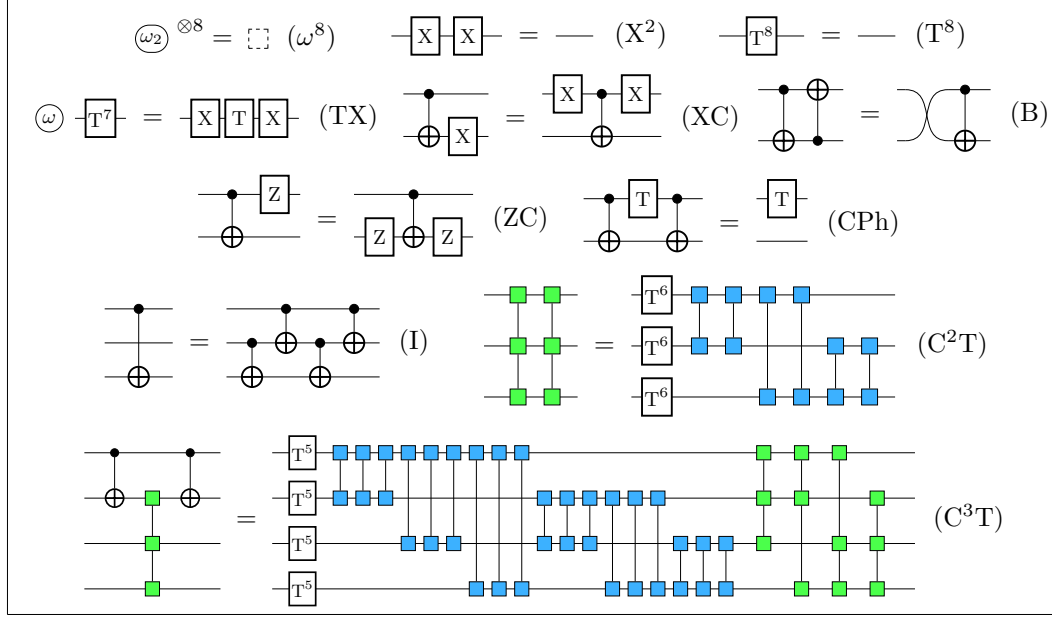
$$\llbracket C \rrbracket = \llbracket E(C) \rrbracket_{\text{src}} \quad \text{and} \quad \llbracket C' \rrbracket_{\text{src}} = \llbracket D(C') \rrbracket. \quad (1)$$

Intuitively, E expresses target syntax in the source language, while D evaluates source syntax back in the target one; (1) says that both translations preserve the intended semantics.

The first syntactic check is that decoding after encoding acts as the identity in the target theory. It suffices to verify this on generators.

Lemma 19. Assume $\mathbf{P}$ is presented by generators Σ and relations $\mathcal{R}$, and that for every non-structural generator $g \in \Sigma$, $\mathbf{P}/\mathcal{R} \vdash D(E(g)) = g$. Then for every circuit C built from Σ , $\mathbf{P}/\mathcal{R} \vdash D(E(C)) = C$.

Proof. By structural induction on C , using that E and D preserve $\circ$ and $\otimes$ and fix structural morphisms. $\blacktriangleleft$



■ **Figure 6** Set of relations CNOTdihe2, where the non-generator gates are defined in ??.

The second check is that decoding respects the *source* axioms, so that any derivation in the source theory can be transported through D .

► **Lemma 20.** *Assume that for every axiom $L = R$ in $\mathcal{R}^{\text{src}}$, $\mathbf{P}/\mathcal{R} \vdash D(L) = D(R)$. Then D induces a PROP morphism $\overline{D} : \mathbf{P}^{\text{src}}/\mathcal{R}^{\text{src}} \rightarrow \mathbf{P}/\mathcal{R}$, and in particular for all circuits $C_1, C_2 \in \mathbf{P}^{\text{src}}$, $\mathbf{P}^{\text{src}}/\mathcal{R}^{\text{src}} \vdash C_1 = C_2 \implies \mathbf{P}/\mathcal{R} \vdash D(C_1) = D(C_2)$.*

Proof. The hypothesis means D identifies both sides of every relation in $\mathcal{R}^{\text{src}}$, so it factors through the quotient. ◀

Once an encoding/decoding pair is fixed on generators, the two lemmas above give a finite checklist: prove $D \circ E = \text{id}$ on generators and check each source axiom under D . The completeness proof is then uniform.

► **Theorem 21.** *Let $\mathbf{P}^{\text{src}}/\mathcal{R}^{\text{src}}$ be complete for $\llbracket \cdot \rrbracket_{\text{src}}$, and let $\mathbf{P}/\mathcal{R}$ be interpreted by $\llbracket \cdot \rrbracket$. Suppose there exists an encoding/decoding pair (E, D) satisfying (1) and the hypotheses of Lemmas 19 and 20. Then $\mathbf{P}/\mathcal{R}$ is complete for $\llbracket \cdot \rrbracket$.*

Proof. Let $C_1, C_2 \in \mathbf{P}$ with $\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket$. By (1) we have $\llbracket E(C_1) \rrbracket_{\text{src}} = \llbracket E(C_2) \rrbracket_{\text{src}}$, hence by completeness of the source presentation $\mathbf{P}^{\text{src}}/\mathcal{R}^{\text{src}} \vdash E(C_1) = E(C_2)$. Applying $\overline{D}$ from Lemma 20 yields $\mathbf{P}/\mathcal{R} \vdash D(E(C_1)) = D(E(C_2))$, and Lemma 19 rewrites both sides to conclude $\mathbf{P}/\mathcal{R} \vdash C_1 = C_2$. ◀

4.3 Fragment-wise completeness

We now apply the transfer scheme to each fragment. The appendices record the source presentations, the translations on generators, and the verification of the two checklists from Lemmas 19 and 20; here we only state the resulting completeness claims.

The source presentations are recalled in ??????????. The encoding/decoding functors are specified on generators in ??. The required generator equalities (Lemma 19) and axiom checks (Lemma 20) are proved in the appendices ?? and ??.

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For qutrit Clifford and Clifford+ CS we first apply the scalar refinements from Section 4.1, so that source and target presentations are interpreted in the same semantic subPROP. We then instantiate Theorem 21 with the encoding/decoding pairs from ??.

- **Lemma 22.** 1. The graphical language $\mathbf{QCirc}_{\text{Cliff}}$ is complete for the semantic subPROP $\mathbf{Qubit}_{\text{Cliff}}$.
 2. The graphical language $\mathbf{QCirc}_{\text{RCliff}}$ is complete for the semantic subPROP $\mathbf{Qubit}_{\text{RCliff}}$.
 3. The graphical language $\mathbf{QCirc}_{\text{CliffT}}$ is complete up to 2 qubits for the semantic subPROP $\mathbf{Qubit}_{\text{CliffT}}$.
 4. The graphical language $\mathbf{QCirc}_{\text{Cliff3}}$ is complete for the semantic subPROP $\mathbf{Qutrit}_{\text{Cliff3}}$.
 5. The graphical language $\mathbf{QCirc}_{\text{CliffCS}}$ is complete up to 3 qubits for the semantic subPROP $\mathbf{Qubit}_{\text{CliffCS}}$.

For the CNOT-dihedral fragment, the known complete equational theory is already formulated as a PROP with the same scalar generator as in our master signature [1]. Thus no scalar refinement or encoding/decoding step is needed. What remains is to check that the axioms we omit from [1] are derivable from our smaller axiom set; this is carried out in Appendix 6.

- **Lemma 23.** The graphical language $\mathbf{QCirc}_{\text{CNOTdih}}$ is complete for the semantic subPROP $\mathbf{Qubit}_{\text{CNOTdih}}$.

5 Minimality

Completeness ensures that our rule sets derive every semantic equality for the corresponding fragment. In this section we ask the complementary question: are these axioms *irredundant*? Concretely, we formalise axiom independence and prove the claimed independence results uniformly via simple separating interpretations.

5.1 Independence and minimality

This subsection fixes the notion of independence used throughout the section, and explains why in our setting it is naturally *arity-local*. We will use the bounded variant to state minimality claims for fragments where we only consider circuits up to a fixed number of wires.

Recall that each fragment $\bullet$ comes with a free PROP $\mathbf{P}_\bullet$ on a sub-signature and a finite rule set $\mathbf{C}_\bullet$ (cf. Section 3). Since all generators are endomorphisms, any derivation of an equality $C_1 = C_2 : n \rightarrow n$ stays inside the hom-set $\mathbf{P}_\bullet(n, n)$.

For a fragment $\bullet$ and an axiom $\rho \in \mathbf{C}_\bullet$, write $\mathbf{QCirc}_\bullet^{-\rho} := \mathbf{P}_\bullet / (\mathbf{C}_\bullet \setminus \{\rho\})$. Thus $\mathbf{QCirc}_\bullet^{-\rho}$ is the presented PROP obtained by removing ρ and closing under congruence.

- **Definition 24.** Let $\bullet$ be a fragment and let ρ be an axiom $C_1 = C_2$ in $\mathbf{C}_\bullet$. We say that ρ is independent (from the other axioms of the same fragment) if $\mathbf{QCirc}_\bullet^{-\rho} \not\models C_1 = C_2$.

The rule set $\mathbf{C}_\bullet$ is minimal (or irredundant) if every axiom $\rho \in \mathbf{C}_\bullet$ is independent.

Operationally, ρ is independent exactly when the equality it states cannot be derived once ρ is removed.

- **Remark 25.** Fix an intended semantics $\llbracket \cdot \rrbracket$ and assume $\mathbf{C}_\bullet$ is sound for $\llbracket \cdot \rrbracket$. Then an axiom $\rho \in \mathbf{C}_\bullet$ is *necessary for completeness* (removing it breaks completeness) if and only if ρ is independent in the sense of Definition 24. Indeed, if ρ were derivable from the other axioms then removing it would not change the induced congruence; conversely, if $\mathbf{C}_\bullet \setminus \{\rho\}$ were still complete and ρ is semantically valid, completeness would force ρ to be derivable.

Because derivations preserve arity, it is natural to truncate a rule set when studying bounded-wire fragments.

► **Definition 26.** For $k \in \mathbb{N}$ set $C_{\bullet, \leq k} := \{\rho \in C_{\bullet} \mid \rho : n \rightarrow n \text{ with } n \leq k\}$. We say $C_{\bullet}$ is minimal up to k wires if each $\rho \in C_{\bullet, \leq k}$ is independent relative to $C_{\bullet, \leq k}$.

► **Remark 27.** Derivations preserve arity (all generators are endomorphisms), so axioms of arity $> n$ are unusable in any derivation inside $P_{\bullet}(n, n)$.

5.2 Separation by alternative interpretations

Our independence proofs are semantic: to show that an axiom is *not* derivable from the others, we exhibit a model that validates all remaining axioms but distinguishes the two sides of the chosen one. The following lemma packages this standard “separating model” argument in our PROP setting.

► **Lemma 28.** Let $\rho : C_1 = C_2$ be an axiom in $C_{\bullet}$. If some strict symmetric monoidal $F : P_{\bullet} \rightarrow \mathcal{D}$ equalizes every axiom in $C_{\bullet} \setminus \{\rho\}$ but satisfies $F(C_1) \neq F(C_2)$, then ρ is independent.

Proof. Otherwise $C_1 = C_2$ would hold in the quotient by $C_{\bullet} \setminus \{\rho\}$ and hence be preserved by F , contradiction. ◀

► **Example 29.** In the qubit Clifford fragment, to separate the axiom (H^2) (i.e. $H^2 = \text{id}$), consider the occurrence functor $? \dashv \boxed{H} : P_{\text{Cliff}} \rightarrow \mathbf{Has}$ that sends $\boxed{H}$ to 1 and all other generators to 0 (and hence maps a circuit to 1 if and only if it contains an H). Then $? \dashv \boxed{H}$ (LHS of (H^2)) = 1 while $? \dashv \boxed{H}$ (RHS of (H^2)) = 0, whereas every other Clifford axiom has either an H on both sides or on neither side. By Lemma 28, (H^2) is independent.

5.3 Separating interpretations used in Figure 7

Each independence claim in Figure 7 is witnessed by a strict symmetric monoidal functor from the free PROP of the fragment to a simple *endomorphism-only* PROP. We use four schematic families, summarized here.

5.3.0.1 Monoid-valued interpretations.

► **Definition 30.** Let $(M, \oplus, 0)$ be a commutative monoid. Define $\mathbf{End}(M)$ to be the PROP with

$$\mathbf{End}(M)(n, m) := \begin{cases} M & n = m, \\ \emptyset & n \neq m, \end{cases}$$

where both $\circ$ and $\otimes$ are interpreted as $\oplus$, identities as 0, and all structural morphisms (in particular swaps) are sent to 0 unless stated otherwise.

► **Definition 31.** We will use the instances

$$\mathbf{Count}_k := \mathbf{End}(\mathbb{Z}_k, +, 0), \quad \mathbf{Has} := \mathbf{End}(\mathbb{B}, \vee, 0) \text{ with } \mathbb{B} = \{0, 1\}.$$

For a finite set G of symbols (generators, and optionally the basic swap $\sigma_{1,1}$, written $\bowtie$ in figures), the label $\#\{G\}_{[k]}$ denotes the unique strict symmetric monoidal functor $P_{\bullet} \rightarrow \mathbf{Count}_k$ sending each symbol in G to 1 and all others to 0 (allowing weights like $\omega \times 2$ when indicated). For a generator g , the label $?g$ denotes the functor $P_{\bullet} \rightarrow \mathbf{Has}$ sending g to 1 and all other generators to 0, so that $?g(C) = 1$ if and only if g occurs in C .

5.3.0.2 Projective substitutions.

► **Definition 32.** Let $\mathbf{PU}_d$ be the PROP of projective unitaries (unitaries modulo global phase), and let π_d denote the quotient map. For each fragment $\bullet$ (with $d = 2$ for qubits and $d = 3$ for qutrits), define the projective interpretation $J_{\bullet}^{\text{proj}} := \pi_d \circ \llbracket \cdot \rrbracket_{\bullet}$.

Given a generator $g : n \rightarrow n$ and any circuit $C : n \rightarrow n$, we denote by $g \mapsto C$ the interpretation obtained from $J_{\bullet}^{\text{proj}}$ by mapping g to $J_{\bullet}^{\text{proj}}(C)$ and leaving all other generators unchanged (so the right-hand side need not itself be a generator).

5.3.0.3 Determinant-phase invariants.

► **Definition 33.** For a qubit unitary U on n wires, define $\delta_n(U) := \arg \det(U)/2^n \in \mathbb{R}/2\pi\mathbb{Z}$.

► **Definition 34.** Fix $k \in \mathbb{N}$. Define $\arg \det_k : \mathbf{P}_{\bullet} \rightarrow \mathbf{End}(\mathbb{R}/2\pi\mathbb{Z}, +, 0)$ on generators $g : n \rightarrow n$ (with $n \leq k$) by $(\arg \det_k)(g) := 2^{k-n} \cdot \arg \det(\llbracket g \rrbracket) \bmod 2\pi$, and extend additively under $\circ$ and $\otimes$. In particular, the basic swap $\sigma_{1,1}$ has $\det(\llbracket \sigma_{1,1} \rrbracket) = -1$ and hence $\arg \det(\llbracket \sigma_{1,1} \rrbracket) = \pi$, so under this scaling convention it contributes $2^{k-2} \cdot \arg \det(\llbracket \sigma_{1,1} \rrbracket) = 2^{k-2}\pi \bmod 2\pi$, i.e. it is π when $k = 2$ and 0 for every $k \geq 3$ (the only cases used later). Accordingly, for $k \geq 3$ we may treat all structural symmetries as mapping to 0.

The factor 2^{k-n} exactly compensates for the $2^m/2^n$ scaling in the tensor-determinant formula, making $\arg \det_k$ a strict symmetric monoidal functor on circuits of arity $\leq k$. Moreover, for $k \geq 3$ it vanishes on all structural wire permutations: on $n \geq 3$ wires these permutations have determinant 1, and the remaining 2-wire case $\sigma_{1,1}$ has phase π but is multiplied by 2^{k-2} and hence becomes 0 mod 2π .

5.4 Minimality results

We now collect the fragment-wise minimality statements. Each independence claim is witnessed by an explicit separating interpretation, recorded (axiom-by-axiom) in Figure 7. The point of the theorem below is to summarise which rule sets are fully settled (all arities) and which are currently proved only under an arity bound.

► **Theorem 35.** The following rule sets are minimal for the indicated classes of circuits (in the sense of Definitions 24 and 26):

1. Cliff_2 is minimal for $\mathbf{P}_{\text{Cliff}}$ -circuits (all arities).
2. RCliff_2 is minimal for $\mathbf{P}_{\text{RCliff}}$ -circuits (all arities).
3. Cliff_3 is minimal up to 2 qutrit wires.²
4. CliffT_2 is minimal up to 1 qubit wire.
5. CliffCS_2 is minimal up to 2 qubit wires.
6. CNOTdihe_2 is minimal for $\mathbf{P}_{\text{CNOTdihe}}$ -circuits (all arities).

Moreover:

- In CliffT_2 , the 2-qubit axioms (CPh), (B) and (CZ) are independent from the remaining axioms.
- In CliffCS_2 , the 3-qubit axiom (I) is independent from the remaining axioms.

Proof. For each axiom ρ covered by the statement, Figure 7 gives a strict symmetric monoidal F_{ρ} that validates the comparison axioms but separates ρ ; apply Lemma 28 (and Definition 26 in the truncated cases). ◀

² The only missing case is separating the 3-qutrit axiom Equation (I); $\arg \det$ does not apply since the controlled- X used there has determinant 1. We conjecture full minimality.

(a) Clifford		(b) Real Clifford		(c) Qutrit Clifford	
Rel.	Interpretation	Rel.	Interpretation	Rel.	Interpretation
(ω^8)	$? \omega$	$(-^2)$	$? \ominus$	(ω^{12})	$? \omega$
(H^2)	$? \neg \boxed{H}$	(H^2)	$? \neg \boxed{H}$	(H^4)	$? \neg \boxed{H}$
(S^4)	$? \neg \boxed{S}$	(Z^2)	$? \neg \boxed{Z}$	(S^3)	$? \neg \boxed{S}$
(CPh)	$? \neg \oplus$	(F)	$\# \{ \omega \}_{[2]}$	(E)	$\# \{ \neg \boxed{H} \}_{[2]}$
(B)	$\# \{ \neg \boxed{X} \}_{[2]}$	(CX^2)	$? \neg \oplus$	(SS')	$\neg \boxed{S} \mapsto \neg \boxed{S} \neg \boxed{X}$
(CZ)	$\# \{ \neg \oplus, \neg \boxed{X} \}_{[2]}$	(B)	$\# \{ \neg \boxed{X} \}_{[2]}$	(CPh)	$? \neg \oplus$
(I)	$\arg \det_3$	(ZC)	$\# \{ \neg \boxed{Z} \}_{[2]}$	(B)	$\# \{ \neg \boxed{X} \}_{[2]}$
		(CF)	$\neg \boxed{Z} \mapsto \neg$	(CZ)	$\# \{ \neg \boxed{S} \}_{[3]}$
		(CZrev)	$\neg \boxed{H} \mapsto \neg$	(KC)	$\# \{ \neg \oplus \}_{[2]}$
		(I)	$\arg \det_3$	(I)	$\neg$

(d) Clifford + T		(e) Clifford + CS		(f) CNOT-dihedral	
Rel.	Interpretation	Rel.	Interpretation	Rel.	Interpretation
(ω^8)	$? \omega$	(ω^8)	$? \omega$	(ω^8)	$? \omega$
(H^2)	$? \neg \boxed{H}$	(H^2)	$? \neg \boxed{H}$	(T^8)	$? \neg \boxed{T}$
(T^8)	$? \neg \boxed{T}$	(S^4)	$? \neg \boxed{S}$	(X^2)	$? \neg \boxed{X}$
(E)	$\# \{ \neg \boxed{H} \}_{[2]}$	(E)	$\# \{ \neg \boxed{H} \}_{[2]}$	(TX)	$\# \{ \omega \}_{[2]}$
(TX)	$\# \{ \neg \boxed{H}, \neg \boxed{T}, \omega \}_{[2]}$	(CPh)	$? \neg \oplus$	(XC)	$\# \{ \neg \boxed{X} \}_{[2]}$
(CPh)	$? \neg \oplus$	(B)	$\# \{ \neg \boxed{X} \}_{[2]}$	(CPh)	$? \neg \oplus$
(B)	$\# \{ \neg \boxed{X} \}_{[2]}$	(XCS)	$\# \{ \neg \boxed{S}, \neg \boxed{H} \}_{[2]}$	(B)	$\# \{ \neg \boxed{X} \}_{[2]}$
(CZ)	$\# \{ \neg \oplus, \neg \boxed{X} \}_{[2]}$	(CSr)	$\neg \boxed{S} \mapsto \neg \boxed{Z}$	(ZC)	$\# \{ \neg \boxed{T}, \omega \times 2 \}_{[8]}$
(CSH)	$\neg$	(CE)	$\neg \boxed{S} \mapsto \neg \boxed{Z}$	(I)	$\# \{ \neg \oplus, \neg \boxed{X} \}_{[2]}$
(HT ²)	$\neg$	(I)	$\arg \det_3$	(C^2T)	$\# \{ \neg \boxed{T}, \omega \times 2 \}_{[4]}$
(HTH)	$\neg$	(SH_k)	$\neg$	(C^3T)	$\arg \det_4$

■ **Figure 7** Separating interpretations used to witness axiom independence. For each relation, the right column gives a strict symmetric monoidal functor into a simple target PROP that equalizes all other axioms of the same fragment but distinguishes the two sides of this relation (via Lemma 28). Entries marked ‘ $\neg$ ’ are relations for which we provide no separating interpretation.

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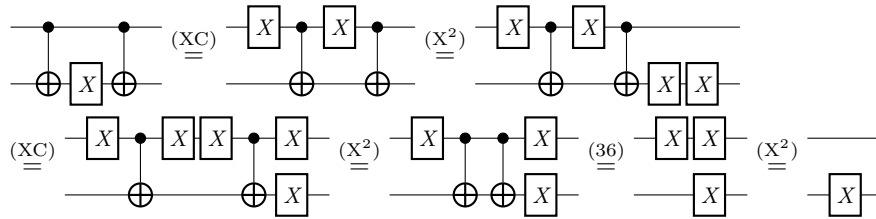
6 CNOT-dihedral circuits

This appendix supplies the fragment-specific proof obligations for the CNOT-dihedral completeness statement in Section 4. Unlike the other fragments, we do not need an encoding/decoding transfer here: the source presentation of [1] already lives in a PROP setting with the same scalar convention as our master signature.

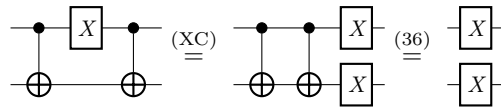
What *does* need to be checked is that our smaller rule set CNOTdihe₂ is conservative with respect to the original one: every axiom of [1] that we omit must be derivable from CNOTdihe₂. The lemmas below provide exactly these derivations, so that the imported completeness theorem applies unchanged.

► **Remark 36.** Axiom (R4) from [1] follows by the same rewriting used in the section for Clifford+T.

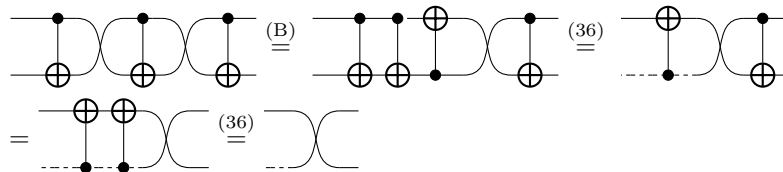
► **Lemma 37.**

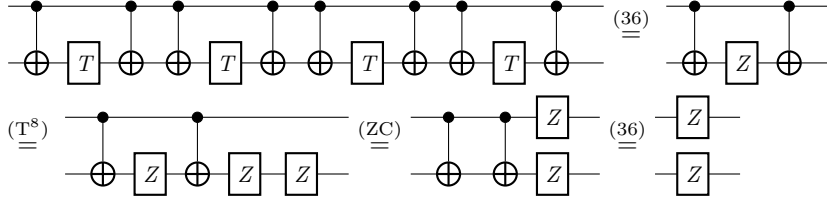


► **Lemma 38.**



► **Lemma 39.**



► **Lemma 40.****7 Scalar refinement**

This appendix justifies the scalar-refinement step used in Section 4.1. The issue is that in a strict (non-projective) semantics, the scalar subgroup generated by a fragment is part of the mathematical object being presented: different choices of primitive phase lead to different semantic subPROPs.

When an imported completeness theorem uses a smaller scalar subgroup than our convention, we refine the *syntax* by adjoining a new scalar generator of the desired order and relating it to the old one. The lemma below proves that this refinement is conservative: it does not collapse any additional equations, and in particular it preserves faithfulness of the interpretation.

7.0.0.1 Instances used in this paper.

The general construction below is instantiated in exactly two places:

- **Qutrit Clifford.** The presentation of [7] works with an order-6 scalar $\textcircled{-\omega}$. To match our strict convention (order 12), we adjoin an order-12 scalar generator $\textcircled{\omega_*}$ and add the relations

$$\textcircled{\omega_*}^{12} = \text{id}_0, \quad \textcircled{\omega_*}^{10} = \textcircled{-\omega}.$$

(This is the concrete “ $\mu_6 \rightarrow \mu_{12}$ ” refinement referenced in the main text.)

- **Clifford+CS.** The presentation of [3] works with an order-4 scalar $\textcircled{i}$. To match our μ_8 convention, we adjoin an order-8 scalar generator $\textcircled{\omega_*}$ and add the relations

$$\textcircled{\omega_*}^8 = \text{id}_0, \quad \textcircled{\omega_*}^2 = \textcircled{i}.$$

(This is the concrete “ $\mu_4 \rightarrow \mu_8$ ” refinement referenced in the main text.)

► **Lemma 41 (Scalar refinement).** *Let $\mathbf{P}^{\text{old}}/\mathcal{R}^{\text{old}}$ be a graphical language equipped with a faithful (i.e. complete) interpretation functor $\llbracket \cdot \rrbracket_{\text{old}} : \mathbf{P}^{\text{old}}/\mathcal{R}^{\text{old}} \rightarrow \mathcal{C}^{\text{old}} \subseteq \mathbf{FdHilb}$.*

Assume that there is a chosen scalar $s : 0 \rightarrow 0$ in $\mathbf{P}^{\text{old}}$ such that:

1. *the scalar group $S(\mathcal{C}^{\text{old}}) := \mathcal{C}^{\text{old}}(0, 0)$ is finite cyclic of order m , generated by $\llbracket s \rrbracket_{\text{old}}$;*
2. *(no hidden phases) for every $n \in \mathbb{N}$ and every $\lambda \in \text{U}(1)$, if $\lambda \text{id}_n \in \mathcal{C}^{\text{old}}(n, n)$, then $\lambda \in S(\mathcal{C}^{\text{old}})$;*
3. *every morphism in $\mathcal{C}^{\text{old}}$ is invertible (equivalently, each endomorphism set $\mathcal{C}^{\text{old}}(n, n)$ is a group under composition).³*

Fix $\ell \geq 1$ and let $\zeta \in \text{U}(1)$ be a root of unity of order $m\ell$. Choose an integer r such that $\zeta^r = \llbracket s \rrbracket_{\text{old}}$. Form the scalar-refined presentation $\mathbf{P}^{\text{old},\#}/\mathcal{R}^{\text{old},\#}$ by adjoining a new scalar generator $\omega : 0 \rightarrow 0$ and adding the two relations $\omega^{m\ell} = \text{id}_0$, $\omega^r = s$.

Let $\mathcal{C}^\#$ be the subPROP of $\mathbf{FdHilb}$ generated by $\mathcal{C}^{\text{old}}$ together with the scalar $\zeta : 0 \rightarrow 0$.

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Then the interpretation functor extends uniquely to a strict symmetric monoidal functor $\llbracket \cdot \rrbracket^\sharp : \mathbf{P}^{\text{old},\sharp} / \mathcal{R}^{\text{old},\sharp} \longrightarrow \mathcal{C}^\sharp$ sending ω to ζ , and $\llbracket \cdot \rrbracket^\sharp$ is faithful.

In particular, the refined language is complete for $\mathcal{C}^\sharp$ and $S(\mathcal{C}^\sharp) \cong \mu_{m\ell}$.

Proof. Soundness. By construction, $\llbracket \cdot \rrbracket^\sharp$ extends $\llbracket \cdot \rrbracket_{\text{old}}$, and the added relations are satisfied because $\zeta^{m\ell} = 1$ and $\zeta^r = \llbracket s \rrbracket_{\text{old}}$.

A categorical fact (centrality of scalars). In any strict symmetric monoidal category, scalars commute with all endomorphisms: for every scalar $a : 0 \rightarrow 0$ and every $f : n \rightarrow n$ one has

$$(a \otimes \text{id}_n) \circ f = f \circ (a \otimes \text{id}_n). \quad (2)$$

Indeed, since 0 is the monoidal unit and the tensor is bifunctorial,

$$(a \otimes \text{id}_n) \circ f = (a \circ \text{id}_0) \otimes (\text{id}_n \circ f) = (\text{id}_0 \circ a) \otimes (f \circ \text{id}_n) = f \circ (a \otimes \text{id}_n),$$

using the interchange law and strictness. As usual, we write $a \otimes f$ for either side of (2).

Scalar extraction normal form. We claim that every morphism $C : n \rightarrow n$ in the refined syntax is provably equal (in $\mathbf{P}^{\text{old},\sharp} / \mathcal{R}^{\text{old},\sharp}$) to

$$C = \omega^a \otimes C', \quad (3)$$

where $0 \leq a < m\ell$ and C' is a morphism built only from the *old* signature (i.e. containing no occurrence of ω). To see this, repeatedly use (2) to slide each occurrence of ω past all other generators and past all compositions/tensors until all ω -factors sit together as a single scalar power. Then reduce the resulting exponent modulo $m\ell$ using $\omega^{m\ell} = \text{id}_0$.

Faithfulness. Let $C_1, C_2 : n \rightarrow n$ be refined circuits such that $\llbracket C_1 \rrbracket^\sharp = \llbracket C_2 \rrbracket^\sharp$. Write them in the normal form (3):

$$C_i = \omega^{a_i} \otimes C'_i \quad (0 \leq a_i < m\ell),$$

and set $U_i := \llbracket C'_i \rrbracket_{\text{old}} \in \mathcal{C}^{\text{old}}(n, n)$. Then the semantic equality becomes $\zeta^{a_1} U_1 = \zeta^{a_2} U_2$.

Using invertibility (3), we can cancel U_2 to obtain

$$\zeta^{a_2 - a_1} \text{id}_n = U_1 \circ U_2^{-1} \in \mathcal{C}^{\text{old}}(n, n). \quad (4)$$

By the no-hidden-phases assumption (2), $\zeta^{a_2 - a_1} \in S(\mathcal{C}^{\text{old}})$. By cyclicity (1), there exists $t \in \{0, \dots, m-1\}$ such that $\zeta^{a_2 - a_1} = \llbracket s \rrbracket_{\text{old}}^t$.

Equivalently, $U_1 = \llbracket s \rrbracket_{\text{old}}^t U_2$, i.e. $\llbracket C'_1 \rrbracket_{\text{old}} = \llbracket s^t \otimes C'_2 \rrbracket_{\text{old}}$.

Since $\llbracket \cdot \rrbracket_{\text{old}}$ is faithful, we conclude that $\mathbf{P}^{\text{old}} / \mathcal{R}^{\text{old}} \vdash C'_1 = s^t \otimes C'_2$.

The refined presentation contains all old relations, so the same equality holds in $\mathbf{P}^{\text{old},\sharp} / \mathcal{R}^{\text{old},\sharp}$. Now use the added relation $\omega^r = s$ to rewrite $s^t = \omega^{rt}$, yielding $C_1 = \omega^{a_1} \otimes C'_1 = \omega^{a_1} \otimes \omega^{rt} \otimes C'_2 = \omega^{a_1 + rt} \otimes C'_2$.

Finally, the semantic identity $\zeta^{a_2 - a_1} = \zeta^{rt}$ and the fact that ζ has order $m\ell$ imply $a_2 - a_1 \equiv rt \pmod{m\ell}$, hence $\omega^{a_1 + rt} = \omega^{a_2}$ using $\omega^{m\ell} = \text{id}_0$. Therefore $C_1 = \omega^{a_2} \otimes C'_2 = C_2$, which proves faithfulness of $\llbracket \cdot \rrbracket^\sharp$.

Scalars. By definition, $S(\mathcal{C}^\sharp)$ is generated by $S(\mathcal{C}^{\text{old}}) = \langle \llbracket s \rrbracket_{\text{old}} \rangle = \langle \zeta^r \rangle$ together with ζ . Since $\zeta^r \in \langle \zeta \rangle$, we have $S(\mathcal{C}^\sharp) = \langle \zeta \rangle \cong \mu_{m\ell}$.

³ In all applications in this paper, $\mathcal{C}^{\text{old}}$ consists of unitaries, so 3 holds automatically.

Interpretation	Cliff	Real Cliff	Qutrit Cliff	Cliff+T	Cliff+CS	CNOT-dih	Arity
$? \omega$	(ω^8)	—	(ω^{12})	(ω^8)	(ω^8)	(ω^8)	0
$? \ominus$	—	$(-^2)$	—	—	—	—	0
$? \boxed{\text{H}}$	(H^2)	(H^2)	(H^4)	(H^2)	(H^2)	—	1
$? \boxed{\text{S}}$	(S^4)	—	(S^3)	—	(S^4)	—	1
$? \boxed{\text{Z}}$	—	(Z^2)	—	—	—	—	1
$? \boxed{\text{T}}$	—	—	—	(T^8)	—	(T^8)	1
$? \boxed{\text{X}}$	—	—	—	—	—	(X^2)	1
$\#\{ \boxed{\text{H}} \}_{[2]}$	(E)	—	(E)	(E)	(E)	—	1
$\#\{ \omega \}_{[2]}$	—	(F)	—	—	—	(TX)	1
$\#\{ \boxed{\text{H}}, \omega \}_{[2]}$	—	—	—	(TX)	—	—	1
$\#\{ \boxed{\text{X}} \}_{[2]}$	—	—	—	—	—	(XC)	1
$\boxed{\text{S}} \mapsto \boxed{\text{S}}\boxed{\text{X}}$	—	—	$(\text{SS}')^{\text{a}}$	—	—	—	1
$? \bigoplus$	(CPh)	(CX^2)	(CPh)	(CPh)	(CPh)	(CPh)	2
$\#\{ \bigwedge \}_{[2]}$	(B)	(B)	(B)	(B)	(B)	(B)	2
$\#\{ \bigoplus, \bigwedge \}_{[2]}$	(CZ)	—	—	(CZ)	—	—	2
$\#\{ \boxed{\text{S}} \}_{[3]}$	—	—	(CZ)	—	—	—	2
$\#\{ \boxed{\text{Z}} \}_{[2]}$	—	(ZC)	—	—	—	—	2
$\boxed{\text{Z}} \mapsto \text{—}$	—	(CF)	—	—	—	—	2
$\boxed{\text{H}} \mapsto \text{—}$	—	(CZrev)	—	—	—	—	2
$\#\{ \bigoplus \}_{[2]}$	—	—	(KC)	—	—	—	2
$\#\{ \boxed{\text{S}}, \boxed{\text{H}} \}_{[2]}$	—	—	—	—	(XCS)	—	2
$\begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \mapsto \begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} \boxed{\text{Z}} \\ \vdots \\ \boxed{\text{Z}} \end{array}$	—	—	—	—	(CSr)	—	2
$\begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \mapsto \begin{array}{c} \bullet \\ \vdots \\ \bullet \end{array} \begin{array}{c} \boxed{\text{Z}} \\ \vdots \\ \boxed{\text{Z}} \end{array}$	—	—	—	—	(CE)	—	2
$\#\{ \boxed{\text{T}}, \omega \times 2 \}_{[8]}$	—	—	—	—	—	(ZC)	2
$\arg \det_3$	(I)	(I)	—	—	(I)	—	3
$\#\{ \bigoplus, \bigwedge \}_{[2]}$	—	—	—	—	—	(I)	3
$\#\{ \boxed{\text{T}}, \omega \times 2 \}_{[4]}$	—	—	—	—	—	(C^2T)	3
$\arg \det_4$	—	—	—	—	—	(C^3T)	4

a) See ??.

■ **Figure 8** Summary of minimality interpretations across all fragments